INCH-POUND

MIL-DTL-44054C
14 September 2011
SUPERSEDING
MIL-B-44054B (GL)
18 June 1985

DETAIL SPECIFICATION

BOTTLE: OIL SAMPLE; WITH CONTAINER, NONAERONAUTICAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers oil sample bottles with containers.

1.2 Classification. The bottles with containers will be of the following types as specified (see 6.2):

Type I - Twelve 3-ounce bottles with caps and container
Type II - Two 3-ounce bottles with caps, two plastic bags and container
Type III - One 3-ounce bottle with cap, one plastic bag and one shipping sack; (container)

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4 or 5 of this specification, whether or not they are listed in this section.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

PPP-B-566 - Boxes, Folding, Paperboard

COMMERCIAL ITEM DESCRIPTIONS

A-A-3174 - Plastic Sheet, Polyolefin

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center Philadelphia, ATTN: DSCP-NASA, 700 Robbins Avenue, Philadelphia, PA 19111-5096 or email to dscpg&ispecomments@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil.

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-23699 - Lubricating Oil, Aircraft Turbine Engine, Synthetic Base, NATO Code Number 0-156

(Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D1974/D1974M - Standard Practice for Methods of Closing, Sealing, and Reinforcing Fiberboard Boxes

ASTM D4727/D4727M - Standard Specification for Corrugated and Solid Fiberboard Sheet Stock (Container Grade) and Cut Shapes

ASTM D5118/D5118M - Standard Practice for Fabrication of Fiberboard Shipping Boxes

(Copies of the above documents are available from ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959 or through their website at http://www.astm.org)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available from www.asq.org or the American Society for Quality, 611 East Wisconsin Avenue, Milwaukee, WI 53202.)

INTERNATIONAL SOCIETY OF BEVERAGE TECHNOLOGISTS

SP – 400 Voluntary Standard

(Copies of this document are available from www.threadspecs.com/assets/Threadspecs/sp400.pdf or the International Society of Beverage Technologists, 14070 Proton Rd., Suite 100, Dallas TX 75244-3601.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.1.1 Containers.

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3.1.1.1 Type I - Shipping box. The shipping box shall be fabricated from corrugated fiberboard material conforming to type CF, class domestic, grade 275 of ASTM D4727/D4727M. The box shall be constructed in accordance with style CSSC of ASTM D5118/D5118M. The manufacturer's joint shall be glued. The bottom flaps shall be closed in accordance with method 2A1 or 2A2 ASTM D1974/D1974M. If staples are used for closure of the bottom flaps, the box shall be provided with a bottom pad made of the same material as the box. Alternatively, box style may be a one-piece folder with top flap tuck and center tuck bottom, or a die-cut, self-erecting folder having a single or double thickness bottom, triple thickness ends and single thickness back and top with a tuck top closure. Each box shall be provided with full-height chipboard partitions that divide the interior of the box into twelve equal sized cells. The box size shall be such that the capped bottles fit into the cells without bulging the partitions nor allow more than 1/8 inch clearance between the bottles and the surfaces of the partition or the box.

3.1.1.2 Type II - Mailing carton. The carton shall be made from paperboard, 5 inches in length, 2 inches in width and 3 inches in depth conforming to variety 1, style II, type D, class A, tongue and tuck of PPP-B-566.

3.1.1.3 Type III - Shipping sack. The shipping sack shall be light weight 6 inches by 10 inches commercial size mailer with pressure sensitive adhesive closure flap.

3.1.2 Shipping box marking. Each type I shipping box shall be marked on two sides with arrows pointing to the top of the box to indicate the position in which the box shall be handled during transit and storage. The word "UP" shall be placed beside the arrows. The letters and arrows shall be at least 1-1/2 inches in height.

3.1.3 Oil sample bottle. The oil sample bottle shall be a wide-mouth cylindrical bottle of one piece construction fabricated of natural, translucent, high density polyethylene with a flame treatment that permits printing ink to adhere. The bottle shall have a threaded finish on top and shall have a rounded shoulder below the bottom of the neck. The open end of the bottle shall have the following characteristics to accommodate a- screw cap closure (see 3.1.5).

a. Continuous type exterior thread in accordance with L style thread profile of SP-400 Voluntary Standard.

b. Flat top surface at the opening that is level and smooth so as to create a leak proof seal with the cap liner.

The bottle shall have a cylindrical shape and shall remain in an upright position when placed on a flat and level surface. The fill capacity of the bottle shall be 95 ± 15 ml when tested as specified in 4.2.1. The exterior surfaces of the bottle shall be smooth, except that impressions from commercial markings, including index slot, and seams from mold marks shall be acceptable but free of flash. The bottle shall be free of cracks, chips, breaks, deformation, sharp edges and burrs. The bottle shall conform to the dimensional requirements specified in table I.

TABLE I. Bottle dimensions.

Inner diameter of opening (mouth and neck)	1.405 inches ± .03 inch
Overall height (with screwcap closure in place)	3.250 inches max.
Wall thickness	0.015 inch min.
Top surface thickness (Brim)	0.060 inch min.
Overall width	2.000 inches max.
Threads	43/400 mm

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3.1.4 Bottle marking. Each bottle shall be printed and blocked as shown on figure 1. Markings shall be applied directly on the bottle by screen printing or any other commercial means using ink that is impervious to oil, diesel fuel, and water. The ink for the printed and blocked areas as shown on figure 1 shall be white and permanently adhered to the bottle. The blocked areas shall be capable of being written on with either ball point pen or pencil. The writing shall remain legible after testing as specified in 4.2.4.

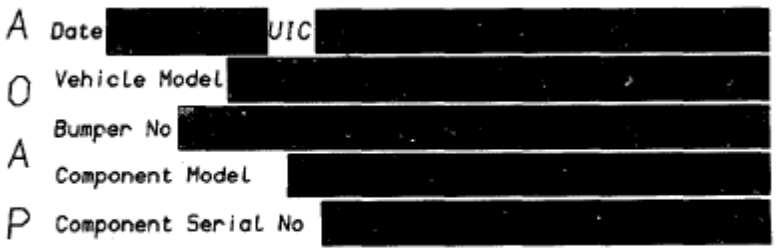


Figure 1. Oil sample bottle marking.

3.1.5 Screwcap. The screwcap shall be a one-piece metal cap of rigid construction having inner edges sufficiently threaded with a continuous type thread to permit closure on the bottle (see 3.1.3). A liner shall be secured to the inside of the cap in such a manner that it does not fall out of the cap, or is not off-center when in the closed position on the bottle. The liner shall be supplied with a facing that is impervious to water, diesel fuel, and oil. The liner shall be of sufficient thickness and resilience to form an effective seal with the flat top surface of the bottle (see 3.1.3). The cap shall not crack, split, break or leak when tested as specified in 4.2.2. The cap shall be free of deformation, cracks, sharp edges, burrs, or protrusions. The cap shall have a finish to resist corrosion.

3.1.6 Bag, plastic. The plastic bags shall be flat, interlocking seal closure with inside dimensions of 6 inches by 6 inches. The plastic film shall conform to type I, class 1, grade A, finish 1, single wall, 0.004 inch thick of A-A-3174.

3.2 Container assembly.

3.2.1 Bottle. Each bottle shall have the screwcap secured in place finger tight.

3.2.2 Type I. When applicable (see 3.1.1.1), a fiberboard pad shall be placed in the bottom of the shipping box. The chipboard partitions shall be set up within the box and twelve bottles, with screw caps secured, shall be placed in each box. The top flaps of CSSC style boxes shall be loosely closed. Alternative style boxes shall-be closed by means of the top flap tuck or the tuck top closure, as applicable.

3.2.3 Type II. Two capped bottles shall be placed in the mailing carton. The two plastic bags shall be neatly folded and placed between the two bottles. Both ends of the carton shall be closed-and tucked.

3.3 Workmanship. The finished oil sample bottles with container shall conform to the quality of product established by this document. The bottles, caps, containers, and bags (when applicable) shall be free from defects which detract from their appearance or may impair their serviceability. All components shall be clean, dry and free of dirt or foreign matter.

4. VERIFICATION

4.1 Conformance inspection. Unless otherwise specified sampling for inspection shall be performed in accordance with ASQ Z1.4. Unless otherwise specified, the Acceptable Quality Limits (AQLs) listed in this section shall be used to establish the sample sizes, however, the acceptance number shall be zero.

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4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document. In addition, the bottles, caps, and boxes shall be examined in accordance with 4.1.1.1 through 4.1.1.3. Bottles, with caps, shall be tested for the characteristics specified in table II. The lot size shall be expressed in units of bottles with caps. The sample unit shall be one bottle with cap. The inspection level shall be S-1 and the acceptable quality limits (AQLs) expressed in terms of defects per hundred units, shall be 2.5.

TABLE II - Component tests.

Characteristic	Requirement reference	Test method
Capacity	3.1.3	4.2.1
Leakage	3.1.3	4.2.2
Pen and pencil marking capability	3.1.4	4.2.3
Legibility after marking	3.1.4	4.2.4

4.1.1.1 Bottle and cap examination. The bottles and caps shall be examined for the defects listed below. The lot size shall be expressed in units of bottles with caps. The sample unit shall be one bottle with cap. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 4.0 for total (major and minor combined) defects.

	Classification	
Defect	Major	Minor
Bottle or cap not free of cracks, chips, or breaks	X	
Bottle or cap not free of distortion	X	
Bottle or cap not free of sharp edges or burrs	X	
Top of bottle not flat or smooth	X	
Bottle not clean, dry, or ready for use	X	
Bottle not free of foreign matter	X	
NOTE: Application of cap to bottle during capping may result in deposit of some small plastic or fiber shavings in the bottle. Presence of a few of these shavings shall not be considered a defect.		
Liner not aligned or secured within screw cap	X	
Bottle not stable when placed on a flat and level surface	X	
Thread on cap or bottle not free of stripping, distortion or discontinuity	X	
Cap not capable of being secured to bottle	X	
Metal caps not free of corrosion	X	
Bottle or cap not free of scratches, blemished or pits		X
Cap not clean		X
Bottle not free of imbedded foreign matter		X
Bottle not free of flash		X

4.1.1.2 Bottle dimensions examination. The bottle shall be examined for conformance to the dimensions specified in table I. Any nonconformance to the specified dimensions shall be classified as a defect. The lot size shall be expressed in units of bottles. The sample unit shall be one bottle. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

4.1.1.3 Box (for bottles) examination. The boxes shall be examined for the defects listed below. The lot size shall be expressed in units of boxes. The sample unit shall be one box. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 4.0 for total (major and minor combined) defects.

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Defect	Classification	
	Major	Minor
Partitions not complete or properly assembled in box	X	
Box not free of deformation	X	
Box not-free of separation	X	
Box not free of tears, cuts, holes., splits, or punctures	X	
Box not capable of being properly closed	X	
Box or partitions not free of moisture	X	
Partition not free of deformation		X
Box or partition not free of dirt or foreign matter		X
Marking on box not complete, legible o present		X

4.2 Methods of inspection. The same bottles comprising the sample size (see 4.1.1) shall be used for all tests specified below. The test bottles shall not be included as part of the contract quantity.

4.2.1 Capacity test. The test shall be conducted at a room temperature of 77 ± 5 °F. The bottles shall be filled with lubricating oil conforming to MIL-PRF-23699 to the transfer bead. The volume of oil used to fill the bottle shall be measured and recorded. Any battle that does not meet the capacity requirement in 3.1.1 shall be a test failure.

4.2.2 Leakage test. The bottles shall be filled to the transfer bead with oil as specified in 4.2.1 and cap closure shall be applied hand tight. The bottles shall be tested for leakage by suspending them in an inverted position for 2 hours at room temperature and then suspended for 4 hours in an oven at 120 ± 10 °F. Any evidence of leakage shall be a test failure.

4.2.3 Pen and pencil marking capability test. The letters and numbers “ABC-123” shall be printed with both ball point pen and pencil on the blocked in ink areas of the bottles. Lack of adhesion of the pen and pencil marking to the white ink shall be a test failure.

4.2.4 Legibility after marking test. The ball point pen marking as performed in 4.2.3 shall be given one firm thumb rub. Illegibility of any letter or number shall be a test failure.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s packaging activities within the Military Service or Defense Agency, or within the military service’s system commands. Packaging data retrieval is available from the managing Military Department’s or Defense Agency’s automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The oil sample container assemblies are used for taking oil samples from surface vehicles and mailing to laboratories for analysis. The oil sample bottle is designed to screw onto the “vampire” oil sampling pump.

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6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, numbers and date of this document.
(b) Type required (see 1.2).
(c) Packaging requirements (see 5.1).

6.3 Subject term (key words) listing.

Analysis
Sampling pump
Surface vehicles
Vampire

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army – GL

Preparing activity:
DLA - IS

(Project 8125-2011-001)

Reviewer activity:
Army – MD

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